

# **BIOINFORMATICS**

## **WEEK 2 – DAY 11**

| Term    | Meaning                                            | Size   |
|---------|----------------------------------------------------|--------|
| Protein | Entire amino acid chain                            | Large  |
| Domain  | Functional unit of a protein                       | Medium |
| Motif   | Small conserved pattern within a domain or protein | Small  |

Book = Protein

Chapter = Domain

Paragraph = Motif

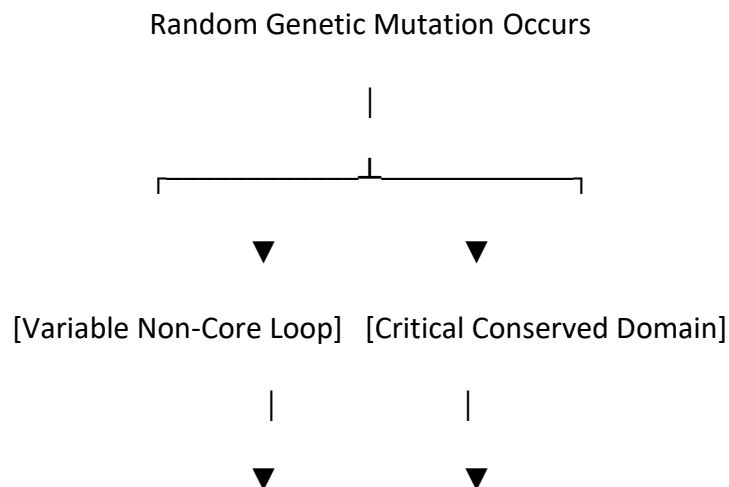
### **Why Are Domains Important?**

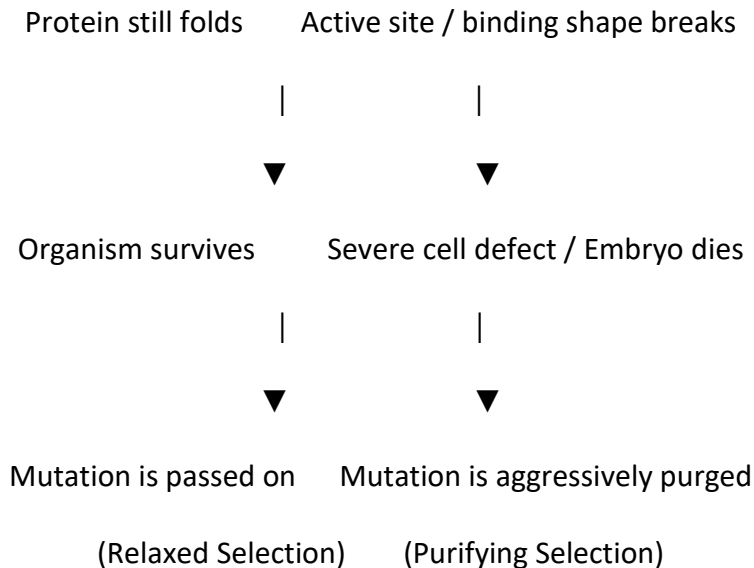
- What does this protein probably do?
- Has evolution preserved this function?
- Does another protein share the same functional module?

Even if two proteins are different overall, sharing the same domain often suggests a related function.

### **The Molecular Mechanics: Why Critical Domains Do Not Change**

To expand your report section with advanced evolutionary mechanics, look at exactly what happens to a protein domain under the hood:





## **Pfam**

It is a comprehensive, globally trusted bioinformatics database registry that catalogs protein families, domains, and functional regions.

Instead of looking at individual amino acid sequences one by one, Pfam groups proteins into evolutionary families based on their shared functional domains.

### **How Pfam Works: Hidden Markov Models (HMMs)**

Pfam does not use simple text-matching lookups (like basic BLAST). Instead, it uses a statistical model called a Hidden Markov Model (HMM).

1. **The Seed:** Curators gather a group of verified sequences that share a known domain.
2. **The Profile:** The HMM analyzes this group to learn which amino acid positions are 100% mandatory and which ones can tolerate mutations.
3. **The Search:** When you upload an unknown sequence, the HMM calculates the probability that your protein fits that functional profile, allowing it to find distant evolutionary relatives that other tools miss.

**Core Classifications** - When you run a search, Pfam breaks down the protein structure into four categories:

- **Domains:** Stable, independent structural blocks that fold on their own to perform a specific task.
- **Families:** Groups of related proteins that share a common evolutionary origin.

- **Repeats:** Short amino acid patterns that repeat back-to-back within a single chain to build a larger structure.
- **Clans:** Superfamilies that group different protein families together if their 3D shapes prove they evolved from the same ancient ancestor.

## **InterPro**

It is a massive bioinformatics database registry managed by the European Bioinformatics Institute (EMBL-EBI). It functions as a "database of databases" by consolidating signatures from 13 different member repositories—including Pfam, SMART, PROSITE, and PRINTS—into a single searchable portal.

Instead of running separate searches across multiple platforms, you paste your sequence into InterPro, and it queries all 13 member registries simultaneously.

## **How InterPro Works: The Power of Integration**

Different databases look at proteins through different analytical lenses. InterPro unifies these methods:

1. **Combines Diverse Methodologies:** It merges databases that look for short, functional amino acid patterns (PROSITE motifs) with databases that build statistical profiles of entire 3D protein shapes (Pfam HMMs).
2. **Eliminates Redundancy:** If Pfam, SMART, and PROSITE all find the exact same binding domain on your protein under different names, InterPro automatically groups them together under a single, unified InterPro Accession Number (IPR...).
3. **Provides Functional Context:** Beyond structural shapes, InterPro maps your protein domains straight to Gene Ontology (GO) terms, telling you exactly what biological process it controls and where it resides in the cell.

## **The InterPro Member Database Ecosystem**

When InterPro scans your sequence, it categorizes the hits based on the member registry that found them:

- **Pfam / SMART:** Isolate independent, structurally stable Domains.
- **PRINTS / PROSITE:** Locate highly specific, short Motifs and Fingerprints (like an enzyme's exact reactive chemical loop).
- **SUPERFAMILY / CATH-Gene3D:** Predict the broader 3D structural folds based on spatial coordinates, even if the sequence itself has heavily mutated.

## **Understanding Domain Architecture**

Imagine a protein has this arrangement:

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DNA-binding | Regulatory | Tetramerization

Each domain contributes a different function.

Remove one domain? The protein may still exist—but its function can change dramatically.

## **Practical Exercise**

**Step 1** - Retrieve the human TP53 protein sequence (FASTA).

Save it as: tp53\_protein.fasta

### **Step 2**

- Open InterPro (or Pfam).
- Paste the protein sequence.
- Run the analysis.

### **Step 3**

Record:

- Number of domains
- Domain names
- Domain positions (start and end amino acid numbers)
- Predicted functions

## **Why Domains Matter in Disease**

Many disease-causing mutations occur within important functional domains.

For example:

- A mutation outside a critical domain may have little effect.
- A mutation inside a DNA-binding domain could prevent a protein from functioning correctly.

Researchers often investigate whether variants fall inside conserved domains to help interpret their significance.